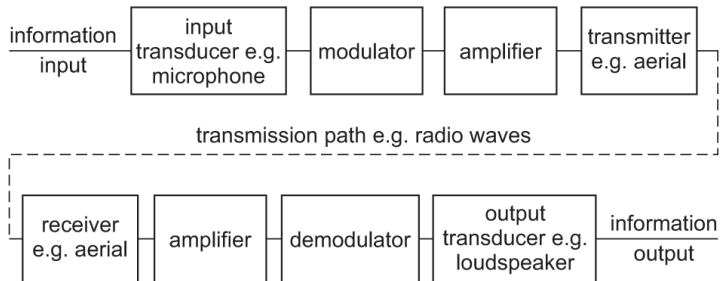


3.13.6 Data communication systems (A-level only)

3.13.6.1 Principles of communication systems (A-level only)

Content

Communication systems, block diagram of 'real time' communication system.



Only the purpose of each stage is required.

Transmission-path media: metal wire, optic fibre, electromagnetic (radio, microwave).

Ground wave, refraction and reflection of sky waves, diffraction of long-wavelength radiation around the Earth's surface.

Satellite systems and typical transmission frequencies.

Students should recognise that up- and down-links require different frequencies so that the receivers are not de-sensed.

Advantages and disadvantages of various transmission media. Students should consider data transmission rate, cost, and security issues.

3.13.6.3 Time-division multiplexing (A-level only)

Content

Basic principles of time-division multiplexing.

3.13.6.4 Amplitude (AM) and frequency modulation (FM) techniques (A-level only)

Content

Principles of modulation; bandwidth.

Carrier wave and information signal.

Details of modulation circuits for modulating a carrier signal with the information signal will not be required.

Graphical representation of both AM and FM modulated signals.

A detailed mathematical treatment is not required.

Students will be expected to identify the carrier frequency and the information frequency from a graph of the variation of signal voltage with time.

Bandwidth requirements of simple AM and FM:

$$\text{bandwidth} = 2f_{\text{M}} \text{ for AM}$$

$$\text{bandwidth} = 2(\Delta f + f_{\text{M}}) \text{ for FM}$$

Data capacity of a channel.

Comparison of bandwidth availability for various media.